MEDICAL EDUCATION

IN THE

UNITED STATES AND CANADA

A REPORT TO

THE CARNEGIE FOUNDATION
FOR THE ADVANCEMENT OF TEACHING

BY

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WITH AN INTRODUCTION BY

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Note on this digest

This digest reproduces, from the 1910 edition of Flexner's report, the title page; part of the Introduction by Henry S. Pritchett, president of the Carnegie Foundation; the opening and the close of Chapter II, where Flexner sets two years of college as the minimum basis of medical education; and ten entries from Part II, in which each school is described under the same headings. The page of the 1910 edition is given in the margin. Omissions are marked by asterisks; running heads and footnotes are left out, and misprints of the scanned edition are corrected. The entries keep the language of 1910.

The report describes no school called "College of Eclectic Medicine." The Eclectic Medical College of New York (pp. 271–272) stands in its place; the Eclectic Medical Institute of Cincinnati (pp. 283–284) would be the other candidate.

The appendix applies Flexner's headings to two schools of 2026. It is a thought experiment, compiled from published sources, which are cited.

Source: Abraham Flexner, *Medical Education in the United States and Canada*, Carnegie Foundation for the Advancement of Teaching, Bulletin Number Four, 1910; scanned edition at the Internet Archive (`https://archive.org/details/medicaleducation00flexiala`).

Introduction

by Henry S. Pritchett

[1910, pp. vii–xi]

THE present report on medical education forms the first of a series of papers on professional schools to be issued by the Carnegie Foundation. The preparation of these papers has grown naturally out of the situation with which the trustees of the Foundation were confronted when they took up the trust committed to them.

When the work of the Foundation began five years ago the trustees found themselves intrusted with an endowment to be expended for the benefit of teachers in the colleges and universities of the United States, Canada, and Newfoundland. It required but the briefest examination to show that amongst the thousand institutions in English-speaking North America which bore the name college or university there was little unity of purpose or of standards. A large majority of all the institutions in the United States bearing the name college were really concerned with secondary education.

Under these conditions the trustees felt themselves compelled to begin a critical study of the work of the college and of the university in different parts of this wide area, and to commend to colleges and universities the adoption of such standards as would intelligently relate the college to the secondary school and to the university. While the Foundation has carefully refrained from attempting to become a standardizing agency, its influence has been thrown in the direction of a differentiation between the secondary school and the college, and between the college and the university. It is indeed only one of a number of agencies, including the stronger colleges and universities, seeking to bring about in American education some fair conception of unity and the attainment ultimately of a system of schools intelligently related to each other and to the ambitions and needs of a democracy.

At the beginning, the Foundation naturally turned its study to the college, as that part of our educational system most directly to be benefited by its endowment. Inevitably, however, the

scrutiny of the college led to the consideration of the relations between the college or university and the professional schools which had gathered about it or were included in it. The confusion found here was quite as great as that which exists between the field of the college and that of the secondary school. Colleges and universities were discovered to have all sorts of relations to their professional schools of law, of medicine, and of theology. In some cases these relations were of the frailest texture, constituting practically only a license from the college by which a proprietary medical school or law school was enabled to live under its name. In other cases the medical school was incorporated into the college or university, but remained an imperium in imperio, the college assuming no responsibility for its standards or its support. In yet other cases the college or university assumed partial obligation of support, but no responsibility for the standards of the professional school, while in only a relatively small number of cases was the school of law or of medicine an integral part of the university, receiving from it university standards and adequate maintenance. For the past two decades there has been a marked tendency to set up some connection between universities and detached medical schools, but under the very loose construction just referred to.

Meanwhile the requirements of medical education have enormously increased. The fundamental sciences upon which medicine depends have been greatly extended. The laboratory has come to furnish alike to the physician and to the surgeon a new means for diagnosing and combating disease. The education of the medical practitioner under these changed conditions makes entirely different demands in respect to both preliminary and professional training.

Under these conditions and in the face of the advancing standards of the best medical schools it was clear that the time had come when the relation of professional education in medicine to the general system of education should be clearly defined. The first step towards such a clear understanding was to ascertain the facts concerning medical education and the medical schools themselves at the present time. In accordance, therefore, with

the recommendation of the president and the executive committee, the trustees of the Carnegie Foundation at their meeting in November, 1908, authorized a study and report upon the schools of medicine and law in the United States and appropriated the money necessary for this undertaking. The present report upon medical education, prepared, under the direction of the Foundation, by Mr. Abraham Flexner, is the first result of that action.

No effort has been spared to procure accurate and detailed information as to the facilities, resources, and methods of instruction of the medical schools. They have not only been separately visited, but every statement made in regard to each detail has been carefully checked with the data in possession of the American Medical Association, likewise obtained by personal inspection, and with the records of the Association of American Medical Colleges, so far as its membership extends. The details as stated go forth with the sanction of at least two, and frequently more, independent observers.

In making this study the schools of all medical sects have been included. It is clear that so long as a man is to practise medicine, the public is equally concerned in his right preparation for that profession, whatever he call himself, — allopath, homeopath, eclectic, osteopath, or whatnot. It is equally clear that he should be grounded in the fundamental sciences upon which medicine rests, whether he practises under one name or under another.

* * *

The significant facts revealed by this study are these:

(1) For twenty-five years past there has been an enormous over-production of uneducated and ill trained medical practitioners. This has been in absolute disregard of the public welfare and without any serious thought of the interests of the public. Taking the United States as a whole, physicians are four or five times as numerous in proportion to population as in older countries like Germany.

(2) Over-production of ill trained men is due in the main to the existence of a very large number of commercial schools, sustained in many cases by advertising methods through which a mass of unprepared youth is drawn out of industrial occupations into the study of medicine.

(3) Until recently the conduct of a medical school was a profitable business, for the methods of instruction were mainly didactic. As the need for laboratories has become more keenly felt, the expenses of an efficient medical school have been greatly increased. The inadequacy of many of these schools may be judged from the fact that nearly half of all our medical schools have incomes below $10,000, and these incomes determine the quality of instruction that they can and do offer. Colleges and universities have in large measure failed in the past twenty-five years to appreciate the great advance in medical education and the increased cost of teaching it along modern lines. Many universities desirous of apparent educational completeness have annexed medical schools without making themselves responsible either for the standards of the professional schools or for their support.

(4) The existence of many of these unnecessary and inadequate medical schools has been defended by the argument that a poor medical school is justified in the interest of the poor boy. It is clear that the poor boy has no right to go into any profession for which he is not willing to obtain adequate preparation; but the facts set forth in this report make it evident that this argument is insincere, and that the excuse which has hitherto been put forward in the name of the poor boy is in reality an argument in behalf of the poor medical school.

(5) A hospital under complete educational control is as necessary to a medical school as is a laboratory of chemistry or pathology. High grade teaching within a hospital introduces a most wholesome and beneficial influence into its routine. Trustees of hospitals, public and private, should therefore go to the limit of their authority in opening hospital wards to teaching, provided only that the universities secure sufficient funds on their side to employ as teachers men who are devoted to clinical science.

In view of these facts, progress for the future would seem to require a very much smaller number of medical schools, better equipped and better conducted than our schools now as a rule are; and the needs of the public would equally require that we have fewer physicians graduated each year, but that these should be better educated and better trained. With this idea accepted, it necessarily follows that the medical school will, if rightly conducted, articulate not only with the university, but with the general system of education. Just what form that articulation must take will vary in the immediate future in different parts of the country. Throughout the eastern and central states the movement under which the medical school articulates with the second year of the college has already gained such impetus that it can be regarded as practically accepted. In the southern states for the present it would seem that articulation with the four-year high school would be a reasonable starting-point for the future. In time the development of secondary education in the south and the growth of the colleges will make it possible for southern medical schools to accept the two-year college basis of preparation. With reasonable prophecy the time is not far distant when, with fair respect for the interests of the public and the need for physicians, the articulation of the medical school with the university may be the same throughout the entire country.

The Proper Basis of Medical Education

Part I, Chapter II

[1910, pp. 20–21, 25–27]

WE have in the preceding chapter briefly indicated three stages in the development of medical education in America, — the preceptorship, the didactic school, the scientific discipline. We have seen how an empirical training of varying excellence, secured through attendance on a preceptor, gave way to the didactic method, which simply communicated a set body of doctrines of very uneven value; how in our own day this didactic school has capitulated to a procedure that seeks, as far as may be, to escape empiricism in order to base the practice of medicine on observed facts of the same order and cogency as pass muster in other fields of pure and applied science. The apprentice saw disease; the didactic pupil heard and read about it; now once more the medical student returns to the patient, whom in the main he left when he parted with his preceptor. But he returns, relying no longer altogether on the senses with which nature endowed him, but with those senses made infinitely more acute, more accurate, and more helpful by the processes and the instruments which the last half-century's progress has placed at his disposal. This is the meaning of the altered aspect of medical training: the old preceptor, be he never so able, could at best feel, see, smell, listen, with his unaided senses. His achievements are not indeed to be lightly dismissed; for his sole reliance upon his senses greatly augmented their power. Succeed as he might, however, his possibilities in the way of reducing, differentiating, and interpreting phenomena, or significant aspects of phenomena, were abruptly limited by his natural powers. These powers are nowadays easily enough transcended. The self-registering thermometer, the stethoscope, the microscope, the correlation of observed symptoms with the outgivings of chemical analysis and biological experimentation, enormously extend the physician's range. He perceives more speedily and more accurately what he is actually dealing with; he knows with far greater assurance the merits or the limitations of the agents which he is in position

to invoke. Though the field of knowledge and certainty is even yet far from coextensive with the field of disease and injury, it is, as far as it goes, open to quick, intelligent, and effective action.

Provided, of course, the physician is himself competent to use the instrumentalities that have been developed! There is just now the rub. Society reaps at this moment but a small fraction of the advantage which current knowledge has the power to confer. That sick man is relatively rare for whom actually all is done that is at this day humanly feasible, — as feasible in the small hamlet as in the large city, in the public hospital as in the private sanatorium. We have indeed in America medical practitioners not inferior to the best elsewhere; but there is probably no other country in the world in which there is so great a distance and so fatal a difference between the best, the average, and the worst.

* * *

Nor do these apparently recondite matters concern only the experimenting investigator, eager to convert patiently acquired knowledge of bacterial and other foes into a rational system of defense against them. For the practical outcome of such investigation is not communicable by rote; it cannot be reduced to prescriptions for mechanical use by the unenlightened practitioner. Modern medicine cannot be formulated in quiz-compends; those who would employ it must trouble to understand it. Moreover, medicine is developing with beneficent rapidity along these same biological and chemical lines. Is our fresh young graduate of five and twenty to keep abreast of its progress? If so, he must, once more, understand; not otherwise can he adopt the new agents and new methods issuing at intervals from each of a dozen fertile laboratories; for rote has no future: it stops where it is. "There can be no doubt," said Huxley, "that the future of pathology and of therapeutics, and therefore of practical medicine, depends upon the extent to which those who occupy themselves with these subjects are trained in the methods and impregnated with the fundamental truths of biology." Now the medical sciences proper — anatomy,

physiology, pathology, pharmacology — already crowd the two years of the curriculum that can be assigned to them; and in so doing, take for granted the more fundamental sciences— biology, physics, and chemistry — for which there is thus no adequate opportunity within the medical school proper. Only at the sacrifice of some essential part of the medical curriculum — and for every such sacrifice the future patients pay — can this curriculum be made to include the preliminary subjects upon which it presumes.

From the foregoing discussion, these conclusions emerge: By the very nature of the case, admission to a really modern medical school must at the very least depend on a competent knowledge of chemistry, biology, and physics. Every departure from this basis is at the expense of medical training itself. From the exclusive standpoint of the medical school it is immaterial where the student gets the instruction. But it is clear that if it is to become the common minimum basis of medical education, some recognized and organized manner of obtaining it must be devised: it cannot be left to the initiative of the individual without greatly impairing its quality. Regular provision must therefore be made at a definite moment of normal educational progress. Now the requirement above agreed on is too extensive and too difficult to be incorporated in its entirety within the high school or to be substituted for a considerable portion of the usual high school course; besides, it demands greater maturity than the secondary school student can be credited with except towards the close of his high school career. The possibility of mastering the three sciences outside of school may be dismissed without argument. In the college or technical school alone can the work be regularly, efficiently, and surely arranged for. The requirement is therefore necessarily a college requirement, covering two years, because three laboratory courses cannot be carried through in a briefer period, — a fortunate circumstance, since it favors the student's simultaneous development along other and more general lines. It appears, then, that a policy that at the outset was considered from the narrow standpoint of the medical school alone shortly involves the abandonment of this point of view in favor of some-

thing more comprehensive. The preliminary requirement for entrance upon medical education must therefore be formulated in terms that establish a distinct relation, pedagogical and chronological, between the medical school and other educational agencies. Nothing will do more to steady and to improve the college itself than its assumption of such definite functions in respect to professional and other forms of special training.

So far we have spoken explicitly of the fundamental sciences only. They furnish, indeed, the essential instrumental basis of medical education. But the instrumental minimum can hardly serve as the permanent professional minimum. It is even instrumentally inadequate. The practitioner deals with facts of two categories. Chemistry, physics, biology enable him to apprehend one set; he needs a different apperceptive and appreciative apparatus to deal with other, more subtle elements. Specific preparation is in this direction much more difficult; one must rely for the requisite insight and sympathy on a varied and enlarging cultural experience. Such enlargement of the physician's horizon is otherwise important, for scientific progress has greatly modified his ethical responsibility. His relation was formerly to his patient— at most to his patient's family; and it was almost altogether remedial. The patient had something the matter with him; the doctor was called in to cure it. Payment of a fee ended the transaction. But the physician's function is fast becoming social and preventive, rather than individual and curative. Upon him society relies to ascertain, and through measures essentially educational to enforce, the conditions that prevent disease and make positively for physical and moral well-being. It goes without saying that this type of doctor is first of all an educated man.

How nearly our present resources — educational and economic — permit us to approach the standards above defined is at bottom a question of fact to be investigated presently. We have concluded that a two-year college training, in which the sciences are "featured," is the minimum basis upon which modern medicine can be successfully taught. If the requisite number of physicians cannot at one point or another be procured at that level, a temporary readjustment may be required; but such an

expedient is to be regarded as a makeshift that asks of the sick a sacrifice that must not be required of them a moment longer than is necessary. Before accepting such a measure, however, it is exceedingly important not to confuse the basis on which society can actually get the number of doctors that it needs with the basis on which our present number of medical schools can keep going. Much depends upon which end we start from.

Medical Schools

Part II: ten of the schools visited

[1910, pp. 199–200]

CONNECTICUT

Population, 1,054,366. Number of physicians, 1424. Ratio, 1: 740. Number of medical schools, 1.

NEW HAVEN: Population, 130,027.

YALE MEDICAL SCHOOL. Organized 1813. An organic part of Yale University.

Entrance requirement: Two years of college work, enforced with such unusual conscientiousness that in passing from the high school to the college standard this year, deficient members of last year's class were refused re-admission. Moreover, the advanced requirement has been actually exacted; out of an entering class of 23, one only is conditioned, — in part of biology. This is probably the lowest percentage of "conditions" that the country affords.

Attendance: 138; 72 per cent from Connecticut.

Teaching staff: 64, 14 being professors. Of these, the teachers in the fundamental branches devote full time to instruction, though they are overworked and without a proper force of assistants; in the clinical branches, the professor of medicine with two assistants is salaried. Small sums are also paid to a few other teachers in the clinical years.

Resources available for maintenance: Fees amounting to $15,325, income from endowment amounting to $10,000, university appropriation of $17,986, making annual budget $43,311.

Laboratory facilities: Well equipped student laboratories for organic chemistry, physiology, and pharmacology; the provision for bacteriology, pathology, and anatomy is less satisfactory. In physiology alone is there internal evidence of progressive activity. The instructors in other branches are overworked, being

called on to carry the routine work of extensive subjects in all their parts without adequate assistance. Under such circumstances, the work, however conscientious, is bound to be limited.

Clinical facilities: The New Haven Hospital, in which the school controls a small number of beds, is very intelligently employed. The obstetrical and gynecological wards, however, are not used for teaching; nor is there a contagious disease pavilion. Postmortems are scarce. Clinical laboratories and teaching-rooms have been improvised close by the hospital; students are thereby enabled to do the clinical laboratory work in connection with assigned cases. Provision is also made there for the independent work of the professors of medicine and surgery.

The dispensary occupies a new and excellent building, but lacks systematic organization as a teaching adjunct. The attendance is adequate; but as the staff service is gratis, it varies greatly in quality in various departments.

Date of visit: January, 1910.

[1910, pp. 234–235]

MARYLAND

Population, 1,319,132. Number of physicians, 2012. Ratio, 1: 658. Number of medical schools, 7.

BALTIMORE: Population, 583,475.

(1) MEDICAL DEPARTMENT OF THE JOHNS HOPKINS UNIVERSITY. Established 1893. An organic university department.

Entrance requirement: The bachelor's degree, representing specific attainments in chemistry, physics, biology, German, and French.

Attendance: 297.

Teaching staff: 112, of whom 23 are professors. All the laboratory teaching is conducted by instructors who give their entire

time to teaching and research; the heads of the clinical departments are salaried teachers attached to the Johns Hopkins Hospital.

Resources available for maintenance: The income from tuition fees is $60,542, that from endowments $19,687, making a total of $80,229. The budget calls for $102,429, not including salaries of the clinical faculty and other items carried by the Johns Hopkins Hospital, which is thus actually an integral part of the medical school. The productive hospital endowments now aggregate $3,632,289, not including the bequests for the Phipps Psychiatric Clinic and the Harriet Lane Johnson Home for Children.

Laboratory facilities: These facilities are in every respect unexcelled. As the institution has been from the beginning on a graduate basis, teaching and research have been always equally prominent in its activities.

Clinical facilities: The Johns Hopkins Hospital and Dispensary provide practically ideal opportunities. The medical staff of the hospital and the clinical faculty of the medical school are identical; the scientific laboratories ranged around the hospital are in close touch with clinical problems, immediate and investigative. The medical school plant is thus an organic whole, in which laboratories and clinics are inextricably interwoven. Recent foundations have greatly augmented the original hospital plant in the direction of psychiatry, pediatrics, and tuberculosis. Three hundred and eighty-five beds under complete control are now available.

The dispensary is largely attended, and is admirably conducted from the standpoint of both public service and pedagogic efficiency.

Date of visit: December, 1909,

[1910, pp. 271–272]

NEW YORK

Population, 8,706,039. Number of physicians, 14,117. Ratio, 1: 617. Number of medical schools, 11, plus 4 postgraduate schools.

NEW YORK: Population, 4,563,604.

(9) ECLECTIC MEDICAL COLLEGE. Organized 1865. An independent school. Entrance requirement: Regents' Medical Student Certificate. Attendance: 96, 84 per cent from New York state.

Teaching staff: 45, 16 being professors, 29 of other grade. No one devotes full time to teaching.

Resources available for maintenance: Mainly fees, amounting to $8311.

Laboratory facilities: The chemical laboratory, adequate for routine teaching, is active. Otherwise the facilities are weak: one room is used for bacteriology, histology, pathology, and clinical microscopy; this with the dissecting-room completes the laboratory outfit. There is no museum; but the school possesses a small collection of models, a materia medica cabinet, a stereopticon, and a fair-sized library, of which the books are mostly not recent.

Clinical facilities: There is no eclectic hospital. Twice weekly three students spend their entire afternoon at the Sydenham Hospital (80 free beds). There are some supplementary clinics of the usual kind. There is a small dispensary in the school building and an outside dispensary is also used. The clinical facilities are utterly inadequate in respect to both extent and control. Date of visit: March, 1909.

[1910, pp. 237–238]

MARYLAND

BALTIMORE: Population, 583,475.

(6) MARYLAND MEDICAL COLLEGE. Organized 1898. An independent institution. Entrance requirement: Nominal.

Attendance: 95. Almost one-half the school is in the senior class. Teaching staff: 39, of whom 21 are professors, 18 of other grade. Resources available for maintenance: Fees, amounting to $7460 (estimated).

Laboratory facilities: The school building is wretchedly dirty. Its so-called laboratories are of the worst existing type: one neglected and filthy room is set aside for bacteriology, pathology, and histology: a few dirty test-tubes stand around in pans and old cigar-boxes. The chemical laboratory is perhaps equal to the teaching of elementary chemistry. The dissecting-room is foul. This description completely exhausts its teaching facilities. There is no museum or library and no teaching accessories of any sort whatsoever. Clinical facilities: The college faculty own and conduct a hospital within a few blocks. It is essentially a private institution, of no great value to students. Less than 50 beds are free.

The dispensary claims a fair attendance.

Date of visit: March, 1909.

[1910, pp. 204–205]

GEORGIA

Population, 2,557,412. Number of physicians, 2887. Ratio, 1: 886. Number of medical schools, 5.

ATLANTA: Population, 118,243.

(3) GEORGIA COLLEGE OF ECLECTIC MEDICINE AND SURGERY. Organized 1877. An independent institution.

Entrance requirement: Nominal.

Attendance: 66.

Teaching staff: 20, of whom 14 are professors and 6 of other grade.

Resources available for maintenance: Fees, amounting to $5655 (estimated).

Laboratory facilities: The school occupies a building which, in respect to filthy conditions, has few equals, but no superiors, among medical schools. Its anatomy room, containing a single cadaver, is indescribably foul; its chemical "laboratory" is composed of old tables and a few bottles, without water, drain, lockers, or reagents; the pathological and histological "laboratory" contains a few dirty slides and three ordinary microscopes.

Clinical facilities: The school is practically without clinical facilities. Its outfit in obstetrics is limited to a tattered manikin.

Nothing more disgraceful calling itself a medical school can be found anywhere. Date of visit: February, 1909.

[1910, p. 240]

MASSACHUSETTS

Population, 3,162,347. Number of physicians, 5,577. Ratio, 1: 567. Number of medical schools, 5.

BOSTON: Population, 629,868.

(1) MEDICAL SCHOOL OF HARVARD UNIVERSITY. Organized 1782. An integral department of Harvard University.

Entrance requirement: The student has a choice between the bachelor's degree or certain definite requirements in science and modern languages representing two years of undergraduate work, provided that in the latter case a higher passing mark is required for graduation. In the present year, out of a first-year class of 62, 60 entered with the bachelor's degree.

Attendance: The total enrolment is 285; about 69 per cent from New England, 53 per cent from Massachusetts.

Teaching staff: 173, of whom 23 are professors; laboratory instructors as a rule devote their entire time to the department.

Resources available for maintenance: The department has an endowment of $3,326,961; the fees are merged in the general income of the school. The annual budget is $251,389, of which $72,037 are derived from tuition fees.

Laboratory facilities: The laboratories are unexcelled in equipment and organization, in respect to both teaching and research.

Clinical facilities: Abundant clinical material is available at the Massachusetts General Hospital, the City Hospital, and elsewhere. But serious restrictions are felt in two directions: (1) While the university is free to secure laboratory men wherever it chooses, it is practically bound to make clinical appointments by seniority, in accordance with the custom prevailing in the hospital which it uses, or to leave its professor without a hospital clinic. In general it follows that the heir to the hospital service is heir to the university chair. In consequence there is a noticeable lack of sympathy between the laboratory and the clinical men. They do not represent the same ideals. There is no question but that an institution of this rank ought to work in the most intimate cooperation with a hospital; and that, if such were the case, the same principles would obtain in selecting clinical teachers as prevail elsewhere in the university. (2) The extent to which hospital material can be utilized is also limited, though less in surgery than in medicine. The teaching is in the main of the demonstrative character. Something more intimate is possible in a limited way with fourth-year students. The hospital services with one exception rotate at the end of periods of four months.

The school is now installing its own dispensary, likely to be of great value in its clinical instruction.

Date of visit: October, 1909.

[1910, p. 202]

DISTRICT OF COLUMBIA

Population, 322,212. Number of physicians, 1231. Ratio, 1: 262. Number of medical schools 3, plus two postgraduate (Army and Navy Medical)

WASHINGTON: Population, 327,044.

(3) HOWARD UNIVERSITY MEDICAL COLLEGE. Organized 1869. An integral part of Howard University.

Entrance requirement: A high school course or its equivalent.

Attendance: 205, most of whom are working their way through. Practically all the students are colored.

Teaching staff: 52, 22 being professors, 30 of other grade.

Resources available for maintenance: The school budget calls for $40,000, of which $26,000 are supplied by student fees, most of the remainder by government appropriation. Though the school has been changed from a night to a day school, the fees raised from $80 to $100, and the admission requirements stiffened, the attendance has nevertheless increased.

Laboratory facilities: The laboratory equipment includes anatomy, pathology, histology, bacteriology, and chemistry. There is no organized museum, though the school possesses a number of specimens, normal and pathological, charts, models, etc.

Clinical facilities: Clinical facilities are provided in the new, thoroughly modern, and adequate government hospital of 278 free beds, with its dispensary, closely identified with the medical school. A pavilion for contagious diseases alone is lacking.

Date of visit: January, 1910.

[1910,
pp. 280–281]

NORTH CAROLINA

Population, 2,142,084. Number of medical schools, 4.

RALEIGH: Population, 20,533.

(4) LEONARD MEDICAL SCHOOL. Colored. Organized 1882. An integral part of Shaw University.

Entrance requirement: Less than four-year high school education.

Attendance: 125.

Teaching staff: 9, of whom 8 are professors, one of other grade.

Resources available for maintenance: Mainly fees and contributions, amounting to $4721, practically all of which is paid to the practitioner teachers.

Laboratory facilities: These comprise a clean and exceedingly well kept dissectingroom, a slight chemical laboratory, and a still slighter equipment for pathology. There are no library, no museum, and no teaching accessories. It is evident that the policy of paying practitioners has absorbed the resources of a school that exists for purely philanthropic objects.

Clinical facilities: These are hardly more than nominal. The school has access to a sixteen-bed hospital, containing at the time of the visit three patients. There is no dispensary at all. About thirty thousand dollars are, however, now available for building a hospital and improving laboratories.

Date of visit: February, 1909.

[1910,
pp. 253–254]

MISSOURI

Population, 3,491,397. Number of physicians, 6323. Ratio 1: 552. Number of medical schools, 12 (plus 1 postgraduate school).

KIRKSVILLE: Population, 8422.

(5) AMERICAN SCHOOL OF OSTEOPATHY. Established 1892 and owned by two individuals.

Entrance requirement: Less than a common school education.

Attendance: 560 (ranging in age from 18 to 54 years).

Teaching staff: 12, with 11 student assistants.

Resources available for maintenance: Fees, amounting to $89,600 (estimated).

Laboratory facilities: These are absurdly inadequate for the number of students, as is likewise the teaching staff. A single room, with a corresponding preparation room, is used as bacteriological and physiological laboratory, a six weeks' course being given by one teacher to successive squads of 32. In the same way separate additional laboratories are provided for chemistry, anatomy, and pathology. Material for pathological demonstration is bought; there is no museum, and no effort is made to save gross material. The dissecting-room is foul. The "professors" in charge of histology, pathology, and bacteriology are senior students.

Clinical facilities: A hospital of 54 beds adjoins, but its work is practically all "surgery;" the ward cases are "occasionally used for clinics. Students witness operations." Obstetrical work is comparatively scanty. There is no other hospital in the town.

A large dispensary is operated. An instructor is at hand the first time the student administers a "treatment;" after that, "only if summoned." A course of twenty lectures on the fallacies of medicine is given, so that the graduate will know why he does not use "drugs." The school is a business in which a large margin of profit is secured by its owners.

The teaching furnished is of the cheapest kind. Its huge income is therefore largely profit.

Date of visit: November, 1909.

[1910, pp. 240–241]

MASSACHUSETTS

BOSTON: Population, 629,868.

(2) TUFTS COLLEGE MEDICAL SCHOOL. Organized 1893. Administratively an integral department of Tufts College, though actual scientific intercourse is not intimate. Entrance requirement: Below an actual four-year high school course, since certificates of uncertain value have been accepted and examinations used cover less than half a high school course. This is the less defensible as 97 percent of the total enrolment come from New England.

Attendance: The attendance is 384; 97 per cent from New England, 80 per cent from Massachusetts.

Teaching staff: 103, of whom 33 are professors. There are five full-time professors and five full-time assistants in pathology, histology, physiology, and chemistry.

Resources available for maintenance: The school relies on its fees, amounting to $59,093, repaying out of them large advances for buildings made out of the general income of the college.

Laboratory facilities: The laboratories are entirely adequate to the teaching work of the school.

Clinical facilities: For medical clinics the school is confined to the Boston City Hospital and the Boston Dispensary, which furnish abundant material under the usual more or less imperfect control. The Carney Hospital provides considerable additional work in surgery; the specialties are cared for in other institutions. The school is thus clinically handicapped in exactly the same way as Harvard, but to a greater degree by reason of its being restricted in its medical clinics to a single municipal hospital and dispensary. Its range of choice in the matter of clinical professors is limited by the same considerations.

Date of visit: October, 1909.

[1910, p. 263]

From the general discussion of New England:

Whether even Boston will or should continue to support two regular schools — Harvard and Tufts — is decidedly doubtful. The enrolment of Tufts, even on the high school basis, is much swollen. The strict enforcement of that standard — and why should it not be enforced? — will greatly reduce the attendance. The inevitable elevation of requirements will still further cut it down. The school has no resources but fees; out of them it cannot possibly provide for the legitimate demands of the near future. It is difficult to see how the department in question can avoid being seriously crippled; for its remarkable prosperity has depended to no slight extent on the inducement held out by low entrance standards. Its only hope of escape is through endowment, first, so that it may develop its laboratories independently of fee income, next, that it may secure its own hospital. Why should such an expensive step be recommended? If New England is in future to be supplied with high-grade doctors, the quickest and cheapest road to that end is to complete Harvard and to develop Yale, rather than to maintain several more or less imperfect institutions.

A Thought Experiment

Appendix

Suppose Flexner's method were applied in 2026 to two schools that train physicians for Brazil from beyond its borders. The visit is imaginary. The entries follow Flexner's headings, but they are compiled from published sources, cited below, and from two premises set by the Abraham-Flexner Group: that the Universidad Central del Paraguay, like Tufts in 1909, admits more students than its laboratory space and clinical facilities can serve; and that UCEBOL has no hospital and does not require two years of college for admission. Where the sources are silent, the entry says so.

One observation comes first. Measured by Flexner's entrance standard, every Brazilian medical school would fail, the best included, because in Brazil, as in the rest of Latin America, students enter medical school directly from secondary school. Pritchett's Introduction allowed for such conditions: in regions where secondary education was still developing, "articulation with the four-year high school would be a reasonable starting-point." The experiment therefore rests on the two standards that do not depend on a country's school system, the laboratory and the hospital, and adds a measure Flexner did not have: the Revalida, the examination Brazil requires of physicians trained abroad before they may practise.

PEDRO JUAN CABALLERO, PARAGUAY: Population, about 116,000. Nine medical schools in 2019.[1]

UNIVERSIDAD CENTRAL DEL PARAGUAY. A private university. Its school at Pedro Juan Caballero opened in 2017; medicine is also taught at Asunción and Ciudad del Este.[2]

[1]Gazeta do Povo, 29 September 2019, https://www.gazetadopovo.com.br/educacao/brasileiros-se-aventuram-em-cursos-precarios-por-diploma-de-medicina/.

[2]The faculty's web page, https://medicina.central.edu.py/, consulted October 2026. The Brazilian press of 2026 calls the university Universidad Interamericana: Midiamax, 6 October 2026.

Entrance requirement: A secondary school certificate. The local press reports no entrance examination for medicine.[3]

Attendance: 4500 in 2019, more than 90 per cent from Brazil.

Teaching staff: Not ascertained.

Resources available for maintenance: Fees; amount not ascertained.

Laboratory facilities: Microscope rooms and a technology center are described in the local press, and graduates speak of laboratories. Their capacity relative to the number of students was not ascertained; the premise of the experiment holds it insufficient.

Clinical facilities: The university's hospital is "under construction," according to its own web page. In 2019 the Regional Hospital of Pedro Juan Caballero had 90 beds and received hundreds of students a day from the city's nine schools.

Examination record: In the first stage of the 2023 Revalida, 1.5 per cent of the school's graduates passed.[4] The university reports better results since then and more than 3000 graduates revalidated; these figures are its own.

Date of visit: None. Compiled from published sources, October 2026.

General considerations. The comparison with Tufts holds in one respect and fails in another. Like Tufts, the school lives on fees, and its numbers depend on an easy admission: Flexner wrote that the prosperity of Tufts "has depended to no slight extent on the inducement held out by low entrance standards." Tufts, however, had laboratories Flexner called "entirely adequate" and the wards of the Boston City Hospital. The hospital of this school is still being built, and the public hospital of its city is shared by nine schools. Flexner's remedy for Tufts was endowment and a hospital of its own. The examination record shows the cost of

[3] Pedrojuaninos News, 27 August 2024, https://pedrojuaninosnews.com/medicina-en-la-ucp-calidad-en-la-ensenanza-y-el-bajo-costo-son-los-mayores-atractivos.html.

[4] Study organized by Mário Scheffer, Faculty of Medicine of the University of São Paulo, from data obtained under Brazil's access to information law; reported 16 August 2023, https://www.eumedicoresidente.com.br/post/revalida-inep-pesquisa-indice-de-aprovados-instituicao-estrangeira.

waiting for them: of every hundred graduates who sat the first stage in 2023, fewer than two passed.

SANTA CRUZ DE LA SIERRA, BOLIVIA.

UNIVERSIDAD CRISTIANA DE BOLIVIA (UCEBOL). A private university. Medicine is taught in six years, the last a twelve-month rotating internship.[5]

Entrance requirement: The *bachiller* (secondary school) diploma, or for foreign students its legalized equivalent. No college work is required.[6]

Attendance: Not ascertained. Its rector spoke in 2019 of "many Brazilian students."[7]

Teaching staff: Not ascertained.

Resources available for maintenance: Fees; amount not ascertained.

Laboratory facilities: The school describes anatomy amphitheaters, experimental operating rooms, laboratories, and rooms with high-fidelity simulators. Not inspected.

Clinical facilities: The school's description of its medical program names a university clinic and a mobile hospital, and no hospital. The internship takes place in hospitals it does not name. In 2019 the school sought an agreement with a Brazilian federal university to prepare its graduates for the Revalida.

Examination record: In the first stage of the 2023 Revalida, 6.2 per cent of the graduates of the "Universidad Cristiana" of Bolivia passed (the study cited above names the school in this way).

Date of visit: None. Compiled from published sources, October 2026.

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[5]UCEBOL, description of the medical program, `https://www.ucebol.edu.bo/carreras/pdf/medicina.pdf`, consulted October 2026.

[6]UCEBOL, admissions, `https://www.ucebol.edu.bo/admisiones.aspx`, consulted October 2026.

[7]Prefeitura de Palmas, 9 August 2019, `https://www.palmas.to.gov.br/core/noticias/comitiva-da-universidade-crista-da-bolivia-conhece-como-funciona-o-atendimento-ao-usuario-do-sus-em-palmas/`.

General considerations. Pritchett set the rule in 1910: "A hospital under complete educational control is as necessary to a medical school as is a laboratory of chemistry or pathology." By that rule, a school whose own description names a clinic and a mobile hospital lacks half of its equipment. Simulation rooms belong to the laboratory; the ward is a separate requirement, and the school meets it in hospitals it neither names nor controls.

Conclusion of the experiment. Flexner judged a school by what it controlled: its standard of admission, its laboratories, its hospital. Applied a century later, the three measures still separate schools, and the Revalida adds a fourth, a test applied after graduation, as the autoclave tests an answer after it is written. Of the two schools examined, one is building the hospital Flexner would have required; the other describes none.